Practical No 01

A) Create an MVC application to perform basic Arithmetic Operations.

Project Name: ArthimeticMVC

Project Type: MVC Application

Program Code:

I. Filename: Index.cshtml

```
@model ArithmeticMVC.Models.ArithmeticModel
@{
    ViewBag.Title = "Arithmetic Operations";
}

<h2>Arithmetic Operation</h2>
@using (Html.BeginForm())
{
    <table>
        <tr>
            <td>Frist Number</td>
            <td>@Html.TextBoxFor(m=>m.Number1)</td>
        </tr>

        <tr>
            <td>Second Number</td>
            <td>@Html.TextBoxFor(m=>m.Number2)</td>
        </tr>

        <tr>
            <td>Operation</td>
            <td>
                @Html.DropDownListFor(m=>m.Operation,
                new SelectList(new[]
                {
                    "Add",
                    "Subtract",
                    "Multiply",
                    "Divide"
                }))
            </td>
        </tr>

        <tr>
            <td></td>
            <td>
                <input type="submit" value="Calculate"/>
            </td>
        </tr>
    </table>
}

@if(!string.IsNullOrEmpty(Model.ErrorMessage))
{
    <h3 style="color:red">@Model.ErrorMessage</h3>
}
else if(Model.Operation!=null)
{
    <h3>Result=@Model.Result</h3>
}
```

II. Filename: HomeController.cs

```
public class HomeController : Controller
{
    public ActionResult Index(ArithmeticModel model)
    {
```

```
        if (ModelState.IsValid)
        {
            switch (model.Operation)
            {
                case "Add":
                    model.Result = model.Number1 + model.Number2;
                    break;
                case "subtract":
                    model.Result = model.Number1 - model.Number2;
                    break;
                case "Multiply":
                    model.Result = model.Number1 * model.Number2;
                    break;
                case "Divide":
                    if (model.Number2 == 0)
                    {
                        model.ErrorMessage = "Divison by zero is not allowed.";
                    }
                    else
                    {
                        model.Result = model.Number1 / model.Number2;
                    }
                    break;
            }
        }
        return View(model);
    }
}
```

III. Filename: ArithmeticModel.cs

```
using System.ComponentModel.DataAnnotations;
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;

namespace ArithmeticMVC.Models
{
    public class ArithmeticModel
    {
        public double Number1 { get; set; }
        public double Number2 { get; set; }
        public string Operation { get; set; }
        public double Result { get; set; }
        public string ErrorMessage { get; set; }
    }
}
```

B) Create an application to print Floyd's triangle till n rows in C#.

Project Name: Floydtriangle

Project Type: Console App

Program Code:

I. Filename: Program.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace FloydTriangle
{
    class Program
    {
        static void Main(string[] args)
        {
            int n, num = 1;

            Console.Write("Enter the number of rows:");
            n = Convert.ToInt32(Console.ReadLine());

            Console.WriteLine("\nFloyd's Triangle:\n");
            for (int i = 1; i <= n; i++)
            {
                for (int j = 1; j <= i; j++)
                {
                    Console.Write(num + " ");
                    num++;
                }
                Console.WriteLine();
            }
            Console.ReadKey();
        }
    }
}
```

C) Write C# code to arrange the name of cities in sorted order. Accept name of 10 cities from the user.

Project Name: CitySorting

Project Type: Console App

I. Filename: Program.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace FloydTriangle
{
    class Program
    {
        static void Main(string[] args)
        {
            int n, num = 1;

            Console.Write("Enter the number of rows:");
            n = Convert.ToInt32(Console.ReadLine());

            Console.WriteLine("\nFloyd's Triangle:\n");
            for (int i = 1; i <= n; i++)
            {
                for (int j = 1; j <= i; j++)
                {
                    Console.Write(num + " ");
                    num++;
                }
                Console.WriteLine();
            }
            Console.ReadKey();
        }
    }
}
```